6.S892 Final Project: Single to Poly-phase Converter

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1 Introduction

I propose a design for a 1.5 kW isolated power converter that outputs 3 phase 120 V_{rms} 60 Hz mains power from 120 V_{rms} 60 Hz mains power. The design uses a multistage architecture shown in Figure 1. The circuit uses a Bridges PFC converter at it's input to keep high power factor and efficiency at high power draw. A high frequency resonant converter is then used to provide isolation across a transformer. And a three phase PWM inverter is used as the output stage to produce an output with low harmonic distortion.

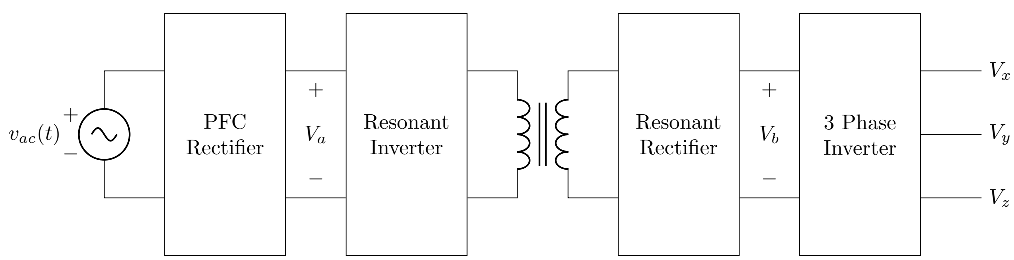

Figure 1: Top Level Schematic

1.1 Requirements

The converter is designed operate with a minimum efficiency above 95% and to pass 1500 W since this is approximately the maximum power a single device will draw from a standard 15 A US outlet ($120\,V \times 15\,A \times 0.8 = 1440\,W$). It is designed to function with input voltages that vary up to ±15% (114 V_{rms} to 138 V_{rms}) in a environment with a temperature ranging from -30 °C to 40 °C.

The converter input should also meet power factor requirement of better that 95%, with a maximum high frequency input current ripple below 100 mA.

2 Bridgeless PFC

Figure 2 shows the schematic of the Bridgless PFC used for this design with specific component values outlined in Table 1. The converter operates with a switching frequency of 500 kHz with a theoretical power factor of ?? and theoretical efficiency of ??.

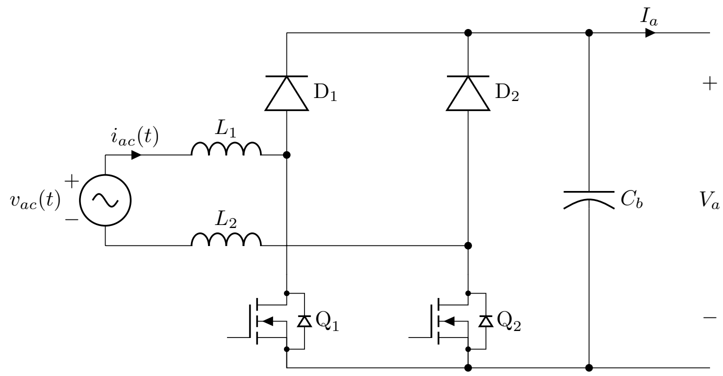

Figure 2: PFC Schematic

Designator	Value	Part
Switching Frequency	500 kHz	–
L_1 and L_2	25 μH	RM14 Core, 1.45 mm gap, 12 turns, 10 AWG
D_1 and D_2	–	TRS12V65H
Q_1 and Q_2	–	IPT60R050G7
C_b	1000uF	3 Series x 4 Parallel UBY2A751MHL

Table 1: PFC Components

The converter takes 120 V AC mains power and converts it into a DC bus with voltage $V_a = 200$ V. Since the converter must tolerate ±15% deviation in voltage the input voltage amplitude range of 144 V to 195 V and a maximum input current amplitude of 20.8 A to 15.37 A.

In periodic steady state the converter will operate with a duty cycle that varies with time as the input voltage changes according to Equation 1. Both switches are controlled by the same signal and switch at the same time, but each will conduct in the reverse direction through their body diode for half the AC cycle.

$$d(t) = \frac{V_a - |v_{ac}(t)|}{V_a} \tag{1}$$

Using this relationship it can be found that, averaging across switching cycles, the MOSFET current is described by Equation 2 and the diode current is described by Equation 3. Importantly for half the cycle the MOSFET is conducting reverse current through it's body diode.

$$\begin{aligned}
\langle i_Q \rangle(t) &= \begin{cases} d(t) \times i_{ac}(t) & 0 < t < \frac{T_{ac}}{2} \\ (1 - d(t)) \times i_{ac}(t) & \frac{T_{ac}}{2} < t < T_{ac} \end{cases} \\
&= \begin{cases} 2P(\frac{1}{V_{ac}}\sin(\omega_{ac}t) - \frac{1}{V_a}\sin^2(\omega_{ac}t)) & 0 < t < \frac{T_{ac}}{2} \\ -\frac{2P}{V_a}\sin^2(\omega_{ac}t) & \frac{T_{ac}}{2} < t < T_{ac} \end{cases}
\end{aligned} \tag{2}$$

$$\langle i_D \rangle(t) = \begin{cases} (1 - d(t)) \times i_{ac}(t) & 0 < t < \frac{T_{ac}}{2} \\ 0 & \text{otherwise} \end{cases} = \begin{cases} \frac{2P}{V_a}\sin^2(\omega_{ac}t) & 0 < t < \frac{T_{ac}}{2} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

2.1 Diodes

Unlike a diodes in a typical rectifier which only switch once an AC cycle, the diodes in a bridgless PFC need to switch at the MOSFET switching frequency. This means it is vital; for them to have a very low reverse recovery time which ends up begin the primary source of loss for silicon diodes. For this reason the design chooses to use SiC diodes which have not reverse recovery and only loose power through their forward voltage drop. This design uses the TRS12V65H diode which have a forward voltage of 1.2 V. This produces a power loss for each individual diode that can be described by Equation 4.

$$
\begin{aligned}
P_{D,cond} &= \frac{1}{T_{ac}} \int_0^{T_{ac}} \langle i_D \rangle(t) V_{D,fwd} \, dt \\
&= \frac{2PV_{D,fwd}}{T_{ac}V_a} \int_0^{\frac{T_{ac}}{2}} \sin^2(\omega_{ac}t) \, dt \\
&= \frac{P}{2V_a} V_{D,fwd} = 4.5\,\text{W}
\end{aligned} \tag{4}
$$

2.2 MOSFETS

In this configuration the MOSFETS will have a maximum drain to Source voltage of $V_a = 200\,\text{V}$ and a maximum current $I_{ac,pk} = 20.8\,\text{A}$. After constraining the search of MOSFETS to only those that can withstand those conditions, the IPT60R050G7 was found to have the lowest theoretical losses of 8.92 W per FET at full load and the lowest expected input voltage amplitude.

2.2.1 Conduction Losses

The FET Conduction loses do not depend on the switching frequency and come from two primary sources: Resistive losses when conducing positive current during half the cycle, and body diode losses when conducting reverse current during the second half of the AC cycle. The body diode losses dominate over the $R_{DS,ON}$ losses. These losses can be calculated with Equation 5, which can also be simplified down to numerical factors for these design parameters to more easily compare FETS as shown in Equation 6. For the IPT60R080G7 FET the maximum expected conduction losses are 4.935 W.

$$
\begin{aligned}
P_{Q,cond} &= \frac{1}{T_{ac}} \int_0^{\frac{T_{ac}}{2}} \langle i_Q \rangle(t)^2 R_{DS,ON} \, dt \; + \; \frac{1}{T_{ac}} \int_{\frac{T_{ac}}{2}}^{T_{ac}} \langle i_Q \rangle(t) - V_{Q,fwd} \, dt \\
&= \frac{2P}{T_{ac}} \int_0^{\frac{T_{ac}}{2}} (\frac{1}{V_{ac}} \sin(\omega_{ac}t) - \frac{1}{V_a} \sin^2(\omega_{ac}t))^2 R_{DS} \, dt + \frac{2PV_{Q,fwd}}{T_{ac}V_a} \int_{\frac{T_{ac}}{2}}^{T_{ac}} \sin^2(\omega_{ac}t) \, dt \\
&= \frac{P^2 R_{DS,ON}}{V_{ac}^2 V_a^2} (V_a^2 - \frac{16V_{ac}V_a}{3\pi} + \frac{3V_{ac}^2}{4}) + \frac{PV_{D,fwd}}{2V_a} \\
&= \frac{P^2}{V_{ac}^2 V_a^2} (V_a^2 - \frac{16V_{ac}V_a}{3\pi} + \frac{3V_{ac}^2}{4}) R_{DS,ON} + \frac{P}{2V_a} V_{D,fwd}
\end{aligned} \tag{5}
$$

$$
P_{Q,cond} = 17.92 \times R_{DS,ON} + 3.75 \times V_{D,fwd} \tag{6}
$$

2.2.2 Dynamic Losses

The FETs second source of losses are the dynamic losses. These include switching losses and capacitive losses which both depend on the switching frequency. Calculations for these losses are shown in Equation 7. Similarly these can be reduces to numerical factors for FET comparison in Equation 9. For the IPT60R080G7 FET the maximum expected dynamic losses are 3.989 W

$$\langle p_{Q,dyn}\rangle(t) = \frac{1}{2} f_{sw} V_a (t_r + t_f) i_{ac}(t) \;+\; \frac{1}{2} C_{oss} V_a^2 f_{sw} \tag{7}$$

$$\begin{aligned} P_{Q,dyn} &= \frac{1}{T_{ax}} \int_0^{\frac{T_{ac}}{2}} \langle p_{Q,sw}\rangle(t)\, dt \\ &= \frac{f_{sw} V_a I_{ac}(t_r + t_f)}{2T_{ax}} \int_0^{\frac{T_{ac}}{2}} \sin(\omega_{ac} t)\, dt \;+\; \frac{1}{2} C_{oss} V_a^2 f_{sw} \\ &= \frac{f_{sw} V_a I_{ac}}{2\pi}(t_r + t_f) \;+\; \frac{1}{2} f_{sw} V_a^2 C_{oss} \end{aligned} \tag{8}$$

$$P_{Q,dyn} = 662 \times f_{sw}(t_r + t_f) + 20000 \times f_{sw} C_{oss} \tag{9}$$

2.3 Inductor

The input inductor for this design is split across the live and neutral sides of the input in order to reduce common mode noise. It has been designed to keep the converter in deep CCM to reduce the high frequency current noise at the input.

2.3.1 Inductance

For a given inductance the input current ripple can be calculated with Equation 10. Dividing by the nominal input current produces yield the ripple ratio at the input shown in Equation 11.

$$\Delta i_L(t) = \frac{d(t) \times v_{ac}(t)}{f_{sw} L} \tag{10}$$

$$\mathcal{R}_L(t) = \frac{d(t) \times v_{ac}(t)}{f_{sw} L \times i_{ac}(t)} = \frac{V_{ac}(1 - \frac{V_{ac}}{V_a}\sin(\omega_{ac} t))}{f_{sw} I_{ac} L} \tag{11}$$

Notably, both the actual ripple amplitude and ripple ratio change throughout the an AC cycle. This reaches a theoretical maximum around where the voltage crosses through zero volts with a maximum ripple ratio given by Equation 12.

$$\mathcal{R}_{L,max} = \frac{V_{ac}}{f_{sw} I_{ac} L} \tag{12}$$

for $\mathcal{R}_{L,max} = 30\%$ and out switching frequency $f_{sw} = 500\,k\text{Hz}$ we get a minimum inductance size of $50\,\mu$H. Split across two inductors each has an inductance of $25\,\mu$H.

2.3.2 Core Selection

In designing the custom magnetics for this stage, we start by selecting a core by using the K_g method shown in Equation 13.

$$K_g \geq \frac{L^2 I_{max}^2 \rho_{cu}}{B_{sat} R k_u} \tag{13}$$

This design requires ans inductance of $L = 25\,\mu\text{H}$ for each inductor which will experience a maximum current of $I_{ac}(1 + \frac{\mathcal{R}}{2}) = 23.92\,\text{A}$. The resistivity of copper in the temperature range the design considers is at it's worst case $\rho_{cu} = 2.09 \times 10^{-8}$ with the final inductor should have a total DC resistance below $R = 1\,\text{m}\Omega$ achieving a packing ratio $k_u = 0.6$. The design will also consider the core to have a saturation limit of $B_{max} = 0.3\text{T}$. Combining these yields a required K_g value above $0.396\,\text{cm}^5$,.

Referencing a table of magnetic design the RM14 core is found to have a $K_g = 0.407 > 0.396$ which should work well for this design.

2.3.3 Inductor Design

The RM14 core has a effective core area of $200\,\text{mm}^2$ and minimum core area or $170\,\text{mm}^2$. The bobbin has a winding area of $107\,\text{mm}^2$ and mean turn length of $71.5\,\text{mm}$.

Since this core is barring current at essentially DC, the limiting factor in the core design is core saturation. The maximum B field experienced in the core will occur where the core area is at a minimum as shown in Equation 14. and from this the minimum number of required turns for the inductor can be found with Equation 15.

$$B_{max} = \frac{\Phi}{A_{min}} = \frac{\mathcal{F}}{\mathcal{R} A_{c,min}} = \frac{I_{max} L}{N A_{c,min}} \tag{14}$$

$$N = \frac{I_{max} L}{B_{max} A_{c,min}} = 11.46 \approx 12\,\text{turns} \tag{15}$$

To form our inductor turns this design uses 10AWG copper wire since this will comfortable keep the current density in the wire below $500\,\frac{\text{A}}{\text{cm}^2}$.

Checking this against the core geometry and our predicted packing density, these coils should fit the core based on Equation 16.

$$\frac{A_W N}{k_u} = 105.2\,\text{mm}^2 < W_A = 107\,\text{mm}^2 \tag{16}$$

Finally, we can gap the inductor to get the required inductance. Since the core permeability is much higher than the permeability of free space, the core reluctance can be ignored for the purposes of this calculation allowing us to use Equation 17 top find the required gap of $1.45\,\text{mm}$.

$$l_g = \frac{N^2 \mu_0 A_c}{L} = 1.45\,\text{mm} \tag{17}$$

2.3.4 Inductor Losses

Since the inductor current is operating in deep CCM the AC component of the current are largely negligible for power loss calculation. The AC component ts of current will produce additional losses to to the proximity and skin effects, ad well as some core loss, but since we are in deep CCM the DC loss will dominate. Therefore a reasonable upper bound for the total loss is twice the DC loss which I will use for this calculation. Notably this assumption is not valid for inductor experiencing a higher magnitude of high frequency ripple such as in the resonant converter which will experience an entirely sinusoidal current with no offset.

The DC resistance in the inductor can be simply calculated based on the bobbin geometry, wire gauge and resistivity of copper as shown in Equation 18

$$R_{DC} = \frac{N l_t \rho_{cu}}{A_W} = 6.93\,\mathrm{m\Omega} \tag{18}$$

This corresponds to a "DC" loss described by Equation 19. Since the total loss is approximated as twice the DC loss each inductor is found to dissipate approximal 3 W

$$P_{L,DC} = \int i_{ac}(t)^2 R_{DC}\, dt = \frac{Iac}{2} R_{DC} = 1.50\,\mathrm{W} \tag{19}$$

2.4 Output Capacitors

The capacitor bank experience both a low frequency current ripple that is twice the AC line frequency, and additional experiences a high frequency current ripple at the PFC switching frequency. It is rather difficult to create a capacitor bank with very high capacity and the ability to handle the large high frequency current ripple from the PFC. Instead This design uses two output capacitors, approximating that one capacitor handles the smoothing the high frequency current ripple while the other acts a bulk storage element so that the converter can produce constant power form the pulsed AC line source.

2.4.1 Required Bulk Capacitance

The bulk capacitance required can be determined by applying a low frequency voltage ripple limit. Since our next stage is a resonate converter the ripple ratio should be relatively small, around 10%. This can be quantified by considering the energy put into and taken out of the capacitor the capacitor by assuming it takes all the low frequency AC ripple. The capacitor power can be approximated by Equation 20 and integrating across it gives the peak to peak change in energy storage show in Equation 21.

$$\langle p_C \rangle(t) = P - p_{ac}(t) = P_{dc} \cos(2\omega_{ac} t) \tag{20}$$

$$\Delta E_C = \frac{P}{\omega_{ac}} \tag{21}$$

Similarly, the change in capacitor energy can be related to the voltage ripple with Equation 22. And by rearranging this an expression can be made for the minimum capacitance given a required voltage maximum ripple ration shown in Equation 23.

$$\Delta E_C = \frac{1}{2} C_B V_{dc} \Delta v_c \tag{22}$$

$$C_B = \frac{P_{dc}}{\omega_{ac} V_a \Delta v_{cb}} = \frac{P_{dc}}{\omega_{ac} V_a^2 \mathcal{R}} \tag{23}$$

For $\mathcal{R} = 10\%$ The converter requires a capacitance of $C_B = 995\,\mu\text{F}$.

2.4.2 Bulk Capacitor Current

Equation 24 approximates the current seen by the bulk capacitor, creating a peak to peak low current ripple of $5.3\,\text{A}_{rms}$ which the bulk capacitor must be capable of withstanding.

$$i_c(t) \approx \frac{p_C(t)}{V_a} = \frac{P_{dc}}{V_a}\cos(2\omega_{ac} t) \tag{24}$$

To achieve this, the design uses 12 $750\,\mu\text{F}$ UBY2A751MHL capacitors in a 3 series 4 parallel configuration. Since each Capacitor is rated for a voltage of $100\,\text{V}$ and $3.67\,\text{A}_{\text{rms}}$ the full bank is $750\,\mu\text{F}$ rated for $300\,\text{V}$ and $14.68\,\text{A}_{\text{rms}}$

2.4.3 Bulk Capacitor Losses

Each UBY2A751MHL capacitor has an equivalent series resistance of $140\,\text{m}\Omega$. Summing across the network the capacitor bank total ESR is $105\,\text{m}\Omega$. The resistive losses here is simple $I^2_{Cb,rms} R_{Cb} = (5.3\,\text{A})^2 (105\,\text{m}\Omega) = 2.95\,\text{W}$.

2.5 Stage Efficiency

The efficiency of the state is worst at maximum power and minim input voltage. So the worst case efficiency can be found by summing the worst case losses from each component and dividing by the maximum power draw as shown in Equation 25 for an efficiency of 97.61%.

$$\begin{aligned}
\%Efficnency &= \left(1 - \frac{2P_D + 2P_Q + 2P_L + P_{Cb}}{P_{max}}\right) \\
&= \left(1 - \frac{2(4.5\,\text{W}) + 2(8.92\,\text{W}) + 2(3\,\text{W}) + (2.95\,\text{W})}{1500\,\text{W}}\right) \\
&= \left(1 - \frac{35.79\,\text{W}}{1500\,\text{W}}\right) = 97.61\% \qquad (25)
\end{aligned}$$

3 Resonant Converter

To provide isolation from the input to the output the circuit uses a high frequency resonant power converter shown in Figure 3 using the components listed in table 2. The converter is an LLC topology since it is a relatively simple structure to build into a simple magnetic component, and it can operate over a wide load range and operates around a frequency of 1 MHz.

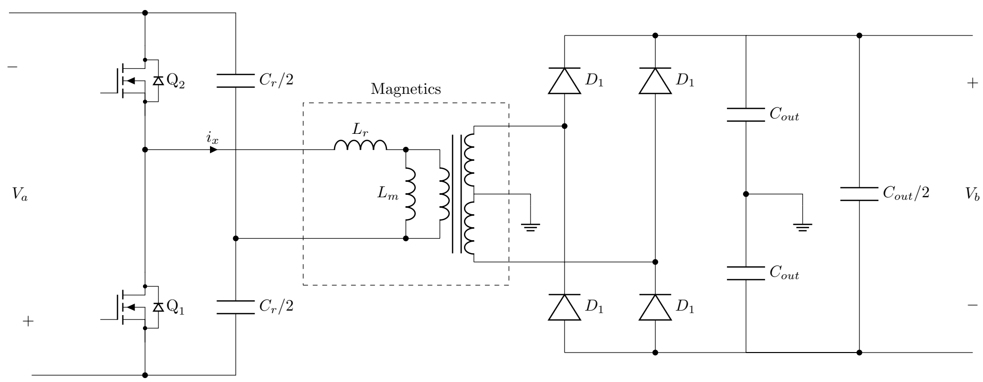

Figure 3: Resonant Converter Schematic

Designator	Value	Part
Switching Frequency	1 MHz	–
Q_1 and Q_2	–	IPL60R105P7
C_r	300 nF	2x C1210W154KCRACTU
Magnetics	$L_r = 86\,\text{nH}$, $L_m = 172\,\text{nH}$	RM12 Core, 9.21 mm core gap with shunt
D_1, D_2, D_3, and D_4	–	TRS12V65H
C_out	75 μF	C4520C0G3F101K200KA and 2x GRM31A5C2J390JW01D

Table 2: Resonant Converter Components

3.1 Half Bridge

Since the converter is designed to run in ZVS soft switching for all loads, the primary deciding factor of the MOSFET is it's on state resistance and body diode forward voltage. It's capacitance and diode recovery time will also influence whether the converter can operate in ZVS, but as long as those values are suitable for the corresponding tank, essentially only the on state resistance and body diode, The body diodes will conduct since the tank will look inductive across it's uprating frequencies. For this reason the design uses the IPL60R105P7 MOSFET for it's reasonably fast switching speeds and low resistance and body diode forward voltage.

Since the resonant tank looks inductive at the driving frequency, a reasonable approximation for the losses in the FETs is to consider $R_{DS,ON}$ loses for a quarter a cycle and body diode losses for a quarter cycle. Since the converter operates in ZVS with soft switching the switching and FET capacitive losses can be ignored. This provides an expression for the losses shown in Equation 26.

$$
\begin{aligned}
P_Q &= \frac{1}{T_{res}}\left(\int_{-\frac{T_{res}}{4}}^{0} i_{res}(t)V_{Q,fwd}\,dt + \int_{0}^{\frac{T_{res}}{4}} i_{res}(t)^2 R_{DS,ON}\,dt\right) \\
&= \frac{1}{T_{res}}\left(\int_{-\frac{T_{res}}{4}}^{0} I_{ppk}\sin(\omega_{res}t)V_{Q,fwd}\,dt + \int_{0}^{\frac{T_{res}}{4}} I_{ppk}^2\sin^2(\omega_{res}t)R_{DS,ON}\,dt\right) \\
&= \frac{I_{ppk}V_{Q,fwd}}{2\pi} + \frac{I_{ppk}^2 R_{DS,ON}}{8} = 17.11\,\text{W} \qquad (26)
\end{aligned}
$$

3.2 Rectifier

The rectifier of this resonant converter is a simple diode bridge. Since the switching frequency is so fast these should be very fast recovery diodes so the design again uses the TRS12V65H diodes which were also used in the PFC. The rectifier losses are relatively simple to calculate considering that two of the four diodes conduct each cycle. The losses per diode are calculated in Equation 27.

$$\begin{aligned} P_D &= \frac{1}{T_{res}} \int_0^{\frac{T_{res}}{2}} i_{res}(t) V_{D,fwd}\, dt \\ &= \frac{1}{T_{res}} \int_0^{\frac{T_{res}}{2}} I_{spk} \sin(\omega_{res} t) V_{Q,fwd}\, dt \\ &= \frac{I_{spk} V_{fwd}}{2\pi} = 2.25\,\mathrm{W} \end{aligned} \tag{27}$$

3.3 Output Capacitance

The rectifier output capacitance can sized with the same method as used for the bulk capacitor of the PFC as shown in Equation 28. Using a ripple ratio of 2%, this means we require a output capacitance of 74.6 nF. as Figure 3 shows this capacitance is split across a network to maintain low ripple relative to ground. The design thus uses one C4520C0G3F101K200KA capacitor between the two power rails, and two GRM31A5C2J390JW01D between the power rails and ground.

$$C_{out} = \frac{P}{\omega_{res} V_b \Delta v_c} = \frac{P}{\omega_{res} V_b^2 \mathcal{R}} \tag{28}$$

3.4 Resonant Tank Design

The process used to design the resonant tank for this design largely follow Texas Instruments Application Note SLUA923. First, the transformer turns ratio is determined using Equation 29. The transformer is designed to have the tank run at resonance with a gain around 1. At this operating point the resonant tank has a current $I_{res,ppk} \approx \frac{\pi P}{V_a} = 23.56$.

$$N_{ps} = \frac{N_p}{N_s} = \frac{V_a - R_{DS,Q1} I_{res,ppk}}{(\frac{V_b}{2} - V_{D1})} = 0.4998 \approx \frac{1}{2} \tag{29}$$

As a clarifying node, this ratio is between the primary and one of the secondary coils. This could also be though of as a 1:4 transformer with the secondary having a center tap.

The second condition that must be met by the is that it continues to operate in ZVS and with soft switching even when no load is present. This requires there to be inductive current to soft charge the capacitor when the FET turns off. Around the operating frequency the magnetizing current will be constant regardless of load and have a phase shift of 90 degrees giving us the current necessary for soft switching. Equation 30 shows the relationship between the converter dead time, switch capacitance, and tank magnetizing current.

$$t_d = \frac{2 C_{oss} V_a}{I_m} \tag{30}$$

It is preferable to keep the dead time short compared to a full switching cycle, approximately 5% so that $t_d = \frac{0.05}{1\,\text{MHz}} = 50\,\text{ns}$. Notably this period is also long enough to avoid shoot through during to rise, fall, and delay times in the MOSFETs switching characteristics. This limits us to a minium required magnetizing current shown in Equation 31.

$$I_m = \frac{2C_{oss}V_a}{t_d} = 5.17\,\text{A} \tag{31}$$

This current requirement in turn creates a minimum required magnetizing inductance given by Equation 32. We will likely want to use a much smaller inductance for the converter so that regulation occurs over a smaller frequency range, but this tells us what is needed to ensure ZVS.

$$L_m \leq \frac{V_a}{I_m f_r} = 38.7\,\mu\text{H} \tag{32}$$

These properties can similarly be used to ensure that the body diode of the FET fully reverse recovers. Before turning off the FET will experience at worst a quarter cycle of positive current, corresponding to at least 250 ns of reverse current delivering at least $5.17\,\mu C$ of charge satisfying the reverse recovery requirements of the FET body diode.

In designing the rest of the tank we are primary concerned with how the converter operates at full load. To do this we can estimate an equivalent resistance seen by the tank as described by Equation 33.

$$R_e = \frac{8N_{ps}^2 V_{out}}{\pi^2 I_{out}} = 5.4\,\Omega \tag{33}$$

A target quality factor of $Q_e = 0.1$ to avoid peaking can be considered to find a target capacitance found in Equation 34 and a target inductance found in Equation 35.

$$C_r = \frac{1}{2\pi Q_e R_e f_r} = 294\,\text{nF} \approx 300\,\text{nF} \tag{34}$$

$$L_r = C_r Q_e^2 R_e^2 = 49.4\,\text{nH} \approx 86\,\text{nH} \tag{35}$$

The resonant capacitor can be simply implemented using two C1210W154KCRACTU 150 nF ceramic capacitors as shown in Figure 3, but the magnetics design is significantly more complex.

Since our input voltage will vary by about 10% with the AC line frequency, we will deign out resonant converter to be able to handle regulating between an output of ±10% across all loads without needing to vary the switching frequency significantly. To do this we will use an inductance ratio of 2, meaning the magnetizing inductance that is twice the resonant inductance. So we will use a mutual inductance of 172 nH.

3.5 Magnetics

Even when it is undesirable a typical transformer will have some leakage and magnetizing inductance which can be represented using simplified magnetic circuit shown in Figure 4.

These reluctance's can be controlled bu gaping the magnetic core for the magnetizing inductance and adding a magnetic shunt in between the primary and secondary coils to control the leakage inductance.

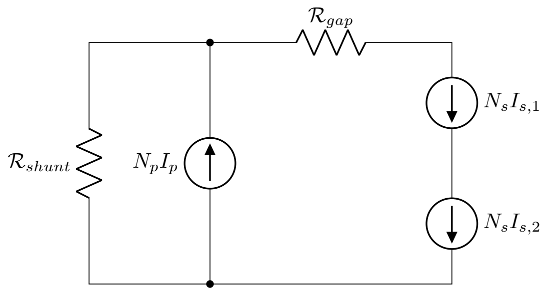

Figure 4: Resonant Converter Magnetic Structure Circuit

The core area product, shown in Equation 36 can be used to estimate a core to use for this design where $f = 1\,\mathrm{MHz}$, $J_0 = 500\,\frac{\mathrm{A}}{\mathrm{cm}^2}$, $k_u = 0.5$, and $S = \frac{I_{res,ppk}V_{res,ppk}}{2} = \frac{127.32\,\mathrm{V}\times 23.56\,\mathrm{A}}{2} = 1500\,\mathrm{W}$. An estimate for B_0 can be found by considering the magnetic performance factor for various magnetic materials. Looking at the materials offered by Ferroxcube, 3F45 is the optimal material having a performance factor of $\mathcal{F} = 58000\,\mathrm{kHz}\times\mathrm{mT}$ corresponding with a $B_0 = 0.058\,\mathrm{T}$.

$$A_C W_A \geq \frac{2S}{\pi f B_0 J_0 k_u} = \frac{3000\,\mathrm{W}}{\pi(1\,\mathrm{MHz})(0.058\,\mathrm{T})(500\,\frac{\mathrm{A}}{\mathrm{cm}^2})(0.5)} = 6585\,\mathrm{mm}^4 \tag{36}$$

Looking again at RM cores, the RM12 core fits the requirements with $A_C W_A = 1.02 \times 10^4\,\mathrm{mm}^4 \geq 6585\,\mathrm{mm}^4$.

Since at high frequencies core loss will dominate, the optimal number of primary windings can be found with Equation 37. With $\beta = 2.51$ and $\beta = 3.06$ being the core materials Steinmetz coefficients.

$$N_{p,opt} = \left(\frac{\beta C_c}{2C_w}\right)^{\frac{1}{\beta+2}} = 3.76 \approx 3 \tag{37}$$

C_c and C_w are given by Equations 38 and 39, respectively. These factors are based on the core geometry and physical constants. The factor F_r is an ac resistance factor that accounts for additional resistance from the skin and proximity effects which can fairly accurately be approximated as 2 by using Litz wire.

$$\begin{aligned} C_c &= V_c k f^{\alpha}\left(\frac{V_{p,pk}}{2\pi f A_c}\right)^{\beta} \\ &= (8320\,\mathrm{mm}^3)(3.28\times 10^{-10})(1\,\mathrm{MHz})^{3.06}\left(\frac{(127.32\,\mathrm{V})}{2\pi\ (1\,\mathrm{MHz})(140\,\mathrm{mm}^2)}\right)^{2.51} \\ &= 48.8 \end{aligned} \tag{38}$$

$$C_w = \frac{4I_{p,pk}^2 \rho_{Cu} l_t F_r}{k_u W_A} = \frac{4(23.56\,\mathrm{A})^2(2.09\times 10^{-8})(61\,\mathrm{mm})(2)}{(0.5)(61\,\mathrm{mm}^2)} = 0.155 \tag{39}$$

This calculation also estimates the total losses in the transformer through Equation 40.

$$P_{tot} = C_c N^{-\beta} + C_w N^2 = 4.49\,\mathrm{W} \tag{40}$$

By considering a maximum current density we can also determine a wire gauge for both the primary coil of 10 AWG and 16 AWG for the secondary, which we can ensure fits in the bobbin window area with room for a shunt using the inequality in Equation 41.

$$\begin{aligned} W_A &\geq \frac{N_p A_{10AWG} + 2N_s A_{16AWG}}{k_u} \\ 73\,\mathrm{mm}^2 &\geq \frac{3(5.26\,\mathrm{mm}^2) + 2\times 6(1.31\,\mathrm{mm}^2)}{0.5} = 63\,\mathrm{mm}^3 \end{aligned} \tag{41}$$

Now that we have a number of turns for our primary winds, and thus secondary winding as well, our core gap can be sized to get the desired mutual inductance in the same manner as done for the PFC following Equation 42.

$$l_g = \frac{N^2 \mu_0 A_c}{L_m} = 9.21\,\mathrm{mm} \tag{42}$$

We can additionally size a piece of shunt material to yield the desired leakage inductance. This shunt will be a cylindrical "tube" of 0.5 mm silicon steal separating the primary and secondary coils with a combined gap of size $l_{g,s}$ at the top and bottom of the core window as claculated with Equation 43.

$$l_{g,s} = \frac{N^2 \mu_0 A_s}{L_r} = \frac{(3)^2(4\pi\times 10^{-7})(\pi \times 19\,\mathrm{mm})}{86\,\mathrm{nH}} = 3.72\,\mathrm{mm} \tag{43}$$

3.6 Stage Efficiency

Summing the total losses This stage dissipated 47.71 W at full power draw. Corresponding to a theoretical stage efficiency of 96.8 %.

4 PWM Inverter

The final stage of the converter is a three phase inverter. These are made up of 3 simple half bridge inverters running in PWM mode with a switch frequency of 250 kHz.

While the converter is designed such that the three phases do not need to be balances, maximum loss for this stage are considered at full output power with a load balanced between all three phases. This yields an average output voltage from each phase given by Equation 44, an average current from each phase given by Equation 45, and a PWM duty cycle shown in Equation 46. The other phases are of course shifted by $2\pi/3$.

$$v_x(t) = V_x \sin(\omega_{ac} t) = (169.7\,\mathrm{V}) \times \sin(2\pi(60\,\mathrm{Hz}) \times t) \tag{44}$$

$$i_x(t) = \frac{2P}{3V_x} \sin(\omega_{ac} t) = (5.89\,\mathrm{A}) \times \sin(2\pi(60\,\mathrm{Hz}) \times t) \tag{45}$$

$$d_x(t) = \frac{v_x(t) + V_x}{V_b} = \frac{V_x \sin(\omega_{ac} t) + V_x}{V_b} = 0.42 \times (\sin(\omega_{ac} t) + 1) \tag{46}$$

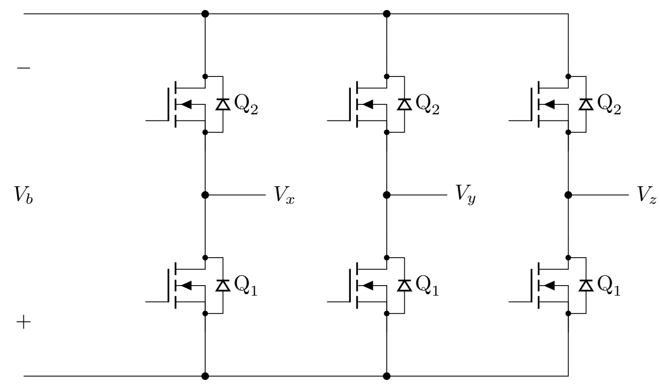

Figure 5: Inverter Schematic

4.1 MOSFETS

This design again uses the same FETs as the PFC, the IPT60R050G7. The losses for this system are also relatively simple to calculate. The conduction losses are shown in Equation 47 and the dynamic losses are shown in Equation 48.

$$\begin{aligned} P_{cond} &= \frac{1}{T_{ac}} \int_0^{T_{ac}} (d(t) \times i_x(t))^2 R_{DS,ON}\, dt \\ &= 0.59\,\mathrm{W} \end{aligned} \tag{47}$$

$$P_{dyn} = \frac{1}{T_{ac}} \int_0^{T_{ac}} \frac{1}{2} i_x(t) v_x(t) f_{sw} (t_r + t_f)\, dt = 0.5625\,\mathrm{W} \tag{48}$$

Summing the calculated losses across all six FETs yields the total losses for this stage of 6.92 W, and a stage efficiency of 99.5%.

5 Overall Efficiency

By summing losses across all stages we can fins that the converter should theoretically have losses of 90.42 W and a theoretical worst case efficiency of 93.97%.

6 Thermal Management

Even with the high effectivity of this converter, since the converter is capable of passing such large power it generates a lot of heat that must be properly handled.

6.1 Passive Components

The passive components are generally large enough that they are capable of radiating their own heat away without need for additional heat sinks. This primarily needs to be consider for the PFC input inductors, the resonant transformer, and the PFC bulk capacitor.

The RM cores luckily have approximate thermal resistance values to ambient, so the temperature rise is easy to calculate. This calculation is shown in Equation 49 for the PFC input inductor and Equation 50 for the resonant transformer. These both keep the cores well under the maximum core temperature of 300 °C at the maximum ambient temperature of 40 °C.

$$\Delta T_{PFC,L} = (3\,\mathrm{W}) \times (19\,^{\circ}\mathrm{C/W}) = 57\,^{\circ}\mathrm{C} \tag{49}$$

$$\Delta T_{res,K} = (4.49\,\mathrm{W}) \times (23\,^{\circ}\mathrm{C/W}) = 103\,^{\circ}\mathrm{C} \tag{50}$$

Verifying this for the capacitor bank is a bit more difficult since these capacitors do not have a well defined thermal characteristics, so experimental testing would be required. However since spread across all the capacitors in the bank the total power dissipated by each capacitor is quite low, we don't predict it will be an issue.

6.2 Active Components

The packages of active components for this design are made for using a single heat sink for the entire design on the back of the PCB and using thermal visa to direct heat. Since the active components are very similar package dimensions the same via array can be used for all of them. Following TI application note AN-2020, an array of 12x8 12 mil vias yields a thermal resistance of about 2.72 °C/W, which fits comfortably in the package footprints. On the back side of the PCB we will use a TG-A1780 Thermal pad and a 9 inch length of a 61340 aluminum extrusion heat sink. This whole design can be assembled into the simplified thermal model shown by the thermal circuit in Figure 6 by combining identical components.

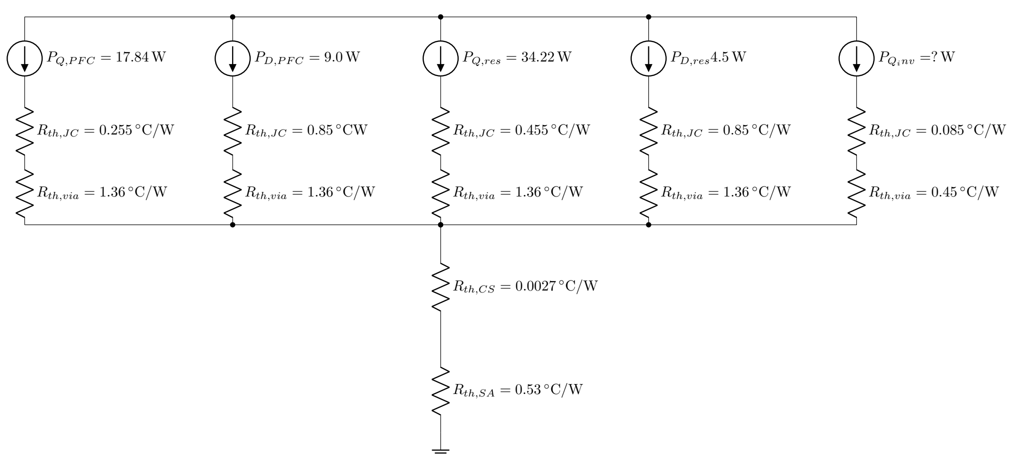

Figure 6: Thermal Circuit Model

Solving this circuit at the maximum ambient temperature, we find that all the components keep a junction temperature below their required 175 °C limit. The FETs in the resonant converter get the hottest, getting a junction temperature as high as 140°C but this is still below the limit. As expected the converter does run quite hot, with a heat sink reaching temperature of 78.4 °C.

7 Conclusions

The design work presented here is a good start, but not comprehensive. Notably, the design is missing input and output filters that would be required to achieve reasonable input and output ripple requirements. Additionally any analysis of the controls needed for this converter and it's transient response are missing. I do plan to complete this work in the future since I want more experience building and controlling high power converters, but this as not completed for the class.

Looking at the design strategies for this converter, there were some poor design decisions. The overall architecture was chosen so that it would touch on a variety of techniques discussed in class, and it is a reasonable design. It is possible it could be improved by using a single stage flyback PFC to get power factor correction and isolation in one stage, though in a well designed system this may be harder on components.

On the topic of "well designed" I don't think this converter is particularly well designed. Particularly the resonant converter has higher losses than I would expect, and more care taken in choosing the FETs may have yielded a much more effective design. Overall I do like the design of this converter and hope to do more with it.